

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	5 MARKS

4.3 – Solution Architecture

Solution Architecture: Comprehensive Analysis and Dietary Strategies with Tableau – A College Food Choices Case Study

The solution architecture for **Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study** is designed to provide an end-to-end analytical framework that collects, processes, analyzes, and visualizes student dietary data to support informed decision-making and healthier eating habits. The architecture begins with a data acquisition layer that gathers information from multiple sources, including student surveys, cafeteria transaction systems, nutritional databases, mobile food-ordering applications, and demographic records. These diverse datasets are integrated through data ingestion mechanisms such as APIs, CSV imports, and database connectors, ensuring seamless data collection from various platforms. The collected data is then stored in a centralized repository using relational databases such as MySQL or PostgreSQL, or cloud-based storage solutions that provide scalability, security, and efficient data management.

The next layer consists of data processing and transformation, where ETL (Extract, Transform, Load) procedures are applied to clean, validate, standardize, and integrate the data. This stage removes duplicate records, handles missing values, and converts raw information into structured datasets suitable for analysis. Advanced data modeling techniques are then employed to create meaningful relationships between variables such as food preferences, nutritional intake, spending behavior, academic workload, and demographic characteristics. Once the data is prepared, Tableau serves as the analytics and visualization layer, enabling the

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creation of interactive dashboards, reports, scorecards, and trend analyses. These visualizations help stakeholders explore key performance indicators (KPIs) such as meal consumption frequency, calorie intake distribution, dietary quality scores, food category preferences, and nutritional deficiencies across different student groups.

The insights generated through Tableau dashboards support a decision-making layer where nutritionists, campus administrators, and policymakers can identify dietary challenges and implement targeted interventions. These interventions may include personalized meal recommendations, healthy cafeteria menu enhancements, nutrition awareness campaigns, budget-friendly meal plans, and wellness programs designed to improve student health outcomes. By integrating data engineering, analytics, visualization, and strategic planning into a unified ecosystem, the proposed solution architecture transforms raw dietary data into actionable intelligence, fostering healthier food choices, improved student well-being, optimized dining services, and a sustainable culture of nutrition-focused decision-making within higher education institutions.

